

Asynchronous Federated Learning based Zero Trust Architecture for the Next Generation Industrial Control Systems[★]

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ABSTRACT

The zero-trust architecture (ZTA) is considered an up-and-coming technology for achieving security in next-generation industrial control systems (ICSs). However, ICSs are intricate and characterized by diverse equipment, cyber-physical integration, dynamic network topologies, and stringent real-time demands, which present formidable challenges for implementing the ZTA. Moreover, as enterprises delve deeper into data sharing to learn advanced business patterns, the risk of data breaches escalates during the digitalization and intelligent transformation journey. To tackle these issues, this article proposed a ZTA for next-generation ICSs based on asynchronous federated deep learning (FDL). Both physical and cyber information are considered for trust evaluation except for subject and object attributes, which can significantly enhance the accuracy of zero-trust decision-making. Moreover, a novel grouping-based asynchronous federated learning algorithm is proposed to reduce the waiting delay of different devices during aggregation, which groups devices with similar computing capabilities and business urgency requirements. In addition, by optimizing the model aggregation, we enhance the model's adaptability to the swift changes in ICSs environments. Through rigorous validation in a real gas pipeline system within our laboratory, we substantiated the efficacy of our proposed ZTA, showcasing its superiority over alternative methodologies.

1. Introduction

Industrial control systems (ICSs), such as Supervisory Control and Data Acquisition (SCADA), Programmable Logic Controller (PLC), Distributed Control System (DCS), Manufacturing Execution System (MES), play a vital role in critical infrastructures like in manufacturing, energy, power, water conservancy, transportation. They are being transformed into next-generation ICSs, with the applications of advanced technologies such as the Industrial Internet of Things (IIoT), Artificial Intelligence (AI), Digital Twin (DT), and next-generation communications. In the next-generation ICSs, enterprises can achieve an improvement in intelligent production management, digital transformation, and efficient inter-enterprise collaboration, thereby significantly enhancing their productivity [1], [2]. However, in transitioning to the next-generation ICSs, the inherent characteristics of existing ICSs pose many challenges to implementing their information security architecture. Firstly, ICSs are designed primarily to meet the business availability and proprietary communication performance, and little consideration is given to security aspects except for

basic error checking. Secondly, the flexible networking and rapid changes in personnel, equipment configurations, and application services increase the difficulty of protecting the system. Furthermore, unlike IT systems, ICS is a complex and time-sensitive system with tight interaction between the cyber world information and physical information. Attacks on ICSs typically start from the cyberworld and penetrate the physical processes to disrupt the production equipment, greatly increasing the difficulties of defense [3], [4], [5].

Traditional perimeter-based defenses, such as intrusion detection and firewalls, aimed at keeping attackers outside the secure perimeter, are no longer sufficient to address the abovementioned challenges. Attackers against ICSs often have a national background and are well-trained. They can exploit vulnerabilities of control devices and software to carry out Advanced Persistent Threats (APTs) attacks. Once attackers breach the defense perimeter, they can easily escalate credential privileges to access the internal network and freely perform cyber-physical attacks on the physical devices of ICSs through lateral movement [6], [7]. Several high-impact incidents have been reported in the past few years, including the infamous "Stuxnet" malware targeting nuclear centrifuges in Iran, the attack on a power grid in Ukraine, and the attack on an oil company in Saudi.

ZTA has recently gained popularity in addressing the abovementioned challenges. It is a dynamic security architecture that significantly differs from traditional perimeter-based defenses. It enables continuous identity authentication and dynamic access control to prevent APT attacks effectively [8]. ZTA has been widely used in various network scenarios, such as mobile communication networks,

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Internet of Things (IoT), wireless sensor networks, etc., and is considered a very promising security architecture for future communication networks [9], [10], [11], [12], [13], [14]. However, very few of these researchers have considered cyber and physical information simultaneously to improve the trust evaluation of ZTA, and most of the existing research has not been validated in a real ICS environment.

Furthermore, the increasing prominence of digital resources with the development of next-generation ICSs has enabled enterprises to gain insights into better business, management, and security models through data sharing[15]. This facilitates business transformation and enhances operational efficiency. However, the implementation of ZTA during data sharing encounters the following problems. Firstly, due to privacy concerns, enterprises are reluctant to share their sensitive business and security data directly with other organizations. Secondly, the isolation of enterprises results in a limited dataset for training ZTA trust assessment models, thereby compromising decision accuracy. Additionally, the similarity in applications and devices among cooperating enterprises increases the likelihood of security risks, which can be easily triggered in new applications. Moreover, the diversity of devices, varying computing capabilities, and proprietary protocols further complicate the application of ZTA in next-generation ICSs.

Federated learning, a novel distributed learning framework, has recently emerged as a promising solution to enhance data sharing security[16]. The primary idea of federated learning is to aggregate the gradients or parameters of the local training models on a centralized node while preserving the privacy-sensitive data on local devices. It has been applied in various sceneries [17], [18], [19], [20] and is typically categorized into asynchronous and synchronous methods. Synchronous federated learning offers high accuracy by using parameters of all local models in each iteration, whereas it incurs a significant communication overhead as all local models must be trained before aggregation [21], [22], [23]. In contrast, asynchronous federated learning allows for immediate aggregation once local models have completed training, thus reducing communication latency and improving aggregation efficiency [24], [25]. It should be noted that the trust evaluation model of ZTA in ICSs must be dynamically updated with rapid environmental changes for accurate decision-making. Therefore, it requires overcoming latency challenges stemming from the varying computational capacity of local nodes in federated learning and enhancing the efficiency of trust evaluation processes.

To cope with the above challenges, especially to adapt to integrating cyber-physical characteristics and the rapidly changing environment, we design a practical federated learning-based ZTA (FedZTA) for the next-generation ICSs to improve aggregation efficiency and maximize decision-making accuracy. The main contributions of this paper are summarized as follows.

- Multi-dimensional properties, including subject, object, cyber, and physical information, are used, and

the enhanced convolutional neural network (CNN) with a multilayer perceptron (MLP) is adopted to improve decision-making accuracy for FedZTA trust evaluation. Specifically, subject information mainly concerns attributes of geographical location, access time, and access frequency; object information considers the importance of devices; cyber information is gathered from network traffic; and physical information focuses on the values of sensors and the status of control devices.

- We construct a novel asynchronous federated learning algorithm by grouping local agents with similar computational abilities and response latency to increase aggregation efficiency. We also verified the effectiveness of our proposed FedZTA through theoretical analysis.
- We perform extensive experiments over real industrial datasets in our lab and a public dataset. Results show that the proposed FedZTA outperforms several existing algorithms in accuracy, especially significantly reducing the average latency.

The rest of the paper is structured as follows: Section 2 describes related work within our research's scope. In Section 3, we outline the methodology of our proposed FedZTA for the next-generation ICSs. Then, comparative experiments are conducted, and a detailed analysis and evaluation of the experimental results are presented in Section 4. In Section 5, we conclude this article and briefly describe opportunities for future improvements to our approach. Finally, in Appendix A, we provided detailed descriptions of the experimental environment and data.

2. Related Work

This section reviews the traditional ZTA and the latest related work of federated learning-based ZTA for ICSs.

2.1. ZTA for ICSs

The National Institute of Standards and Technology (NIST) Special Publication 800-207 gives the definition of Zero Trust (ZT) and outlines the logical components of ZTA [26]. ZTA indicates a new cybersecurity paradigm, shifting focus from perimeter defenses to users, assets, and resources. This approach emphasizes principles such as least privilege, granular access control, and strict policy enforcement to ensure precise access control and prevent unauthorized attacks. Recent efforts have been directed towards enhancing enterprise security through the implementation of ZTA [27], [28], [29], with a particular focus on Industrial Control Systems (ICS) due to their critical national significance [30], [31], [32]. Noteworthy contributions have been made in this area. The work in [30] has devised a novel ZTA named "GAZETA" for defense against lateral movement in 5G IoT Networks. Markov games with one-sided information are utilized to capture dynamic interactions between a potential attacker and the defender. Bayesian

updates are used to perform trust evaluation, and a moving-horizon algorithm is proposed to improve cyber robustness and resilience in the face of unexpected environmental changes. In [31], Lei et al. proposed a three-step ZTA for wireless IIoT building from security zone formation, device authentication, and cryptographic negotiation. Especially artificial noise precoding, physical fingerprint, and physical layer-based key distribution are used to enhance the security of the wireless IIoT. Another work in [32] also develops a ZTA dedicated to industrial cyber-physical systems (ICPS) to model cross-layer penetration. The multi-layer access control engine is constructed to evaluate each component's trust scores, considering each component's different inherent security levels and their cross-layer impact. Besides, the integrated optimizer is proposed to improve efficiency in optimizing access policies.

2.2. Federated learning

In 2016, Google introduced federated learning and put forth FedAvg, a method for federated deep learning through iterative model averaging [33]. In [34], Hu et al. proposed a new algorithm, FedMgda+, viewing federated learning as a multi-objective optimization challenge to ensure fairness among clients. The algorithm focused on converging to a Pareto smoothing solution to avoid sacrificing all participating clients' performances. In the work [21], Chai et al. proposed a hierarchical federated learning system named TiFL that divides clients into different layers based on their training performance. TiFL selects clients from the same layer in each training round to mitigate the problem of reduced aggregation efficiency caused by heterogeneity in resources and data volume. In another work, Karimireddy et al. proposed the SCAFFOLD algorithm, which utilizes a control variable to correct the model walk-off issues during local update [22]. In addition, Xie et al. [24] proposed FedAsync to improve the flexibility and scalability of federated learning. FedAsync also expands asynchronous aggregation for federated averaging and gives a way to deal with outdated models. To reduce the training time and communication resources, Wang et al. introduced the FedCH algorithm by constructing special group topologies and performing hierarchical aggregation [35]. Nguyen et al. [25] proposed a buffered asynchronous aggregation method, FedBuff, to control the number of clients participating in the aggregation by setting a buffer.

2.3. Federated learning-based ZTA for ICSs

Federated learning has been proven efficient in solving the privacy leakage problem in data sharing between interconnected enterprises for the next-generation ICSs. Xie et al. in [12] proposes a comprehensive implementation ZTA for IIoT wireless transmission nodes. Federated learning is utilized to achieve zero-trust rule training, and blockchain technology is employed to aggregate on-chain. The proposed architecture can enhance the accuracy and availability of zero-trust rules while safeguarding IIoT nodes' security. In [36], Ali et al. utilize federated learning to pool computational resources from edge nodes for zero-trust rule

Table 1
Summary of Main Notations

Symbol	Definition
m	The number of agents participating in federated learning
g	The number of groups
N_i	The number of agents in group i
l	The number of iterations for local training
r	The number of global aggregation rounds
L_k	The response latency of agent k
T	Maximum waiting delay during group partitioning
ω	The model parameter
$F(\omega')$	The global loss function after t global aggregations
$F(\omega)$	The local loss function after l iterations for local training

training to address the problem of a single IIoT node with insufficient computational resources not completing model training independently. Each node obtains a local model gradient based on local data and converges gradients to build a global industrial control zero-trust model. In [37], Cheng et al. propose a ZTA for ICSs with a task allocation system optimized by federated learning. It allocates simple computing tasks to devices with scarce resources and assigns more computing tasks to devices with abundant resources. This approach enables all devices to obtain reasonable positions in the decision-making model and improves compatibility among heterogeneous nodes.

3. Proposed FedZTA

This section focuses on the proposed FedZTA system, and the overall structure is shown in Fig.1. The details of the structure are presented in the following three subsections 3.1, 3.2, and 3.3. The operational process of FedZTA is described in subsection 3.4. For clarity, we provide an overview of the key notations employed in this section in Table 1.

3.1. Multi-dimensional information sources

To meet the security requirements of the next-generation ICSs with low-latency and high-concurrency environments, ZTA requires highly accurate trust evaluation solutions. Traditional methods based on subject and object information (e.g., passwords or two-factor authentication) are no longer sufficient for ICSs [38]. Therefore, we expand the information sources for ZTA to adapt the characteristics of ICSs. Specifically, we designed four dimensions of information sources for the Trust Evaluation System (TES): the traditional subject and object information, the cyber information, and the physical information.

- **Subject information:** Subject information refers to information about the entity or user making the access request, marked as $\mathbf{Sub} = \{s_1, s_2, \dots, s_{ns}\}$, where s is a column vector representing an attribute of the subject such as identity, operating system version, geolocation credibility, etc. It provides the necessary context and basis for trust evaluation. By analyzing subject information, the system can assess the

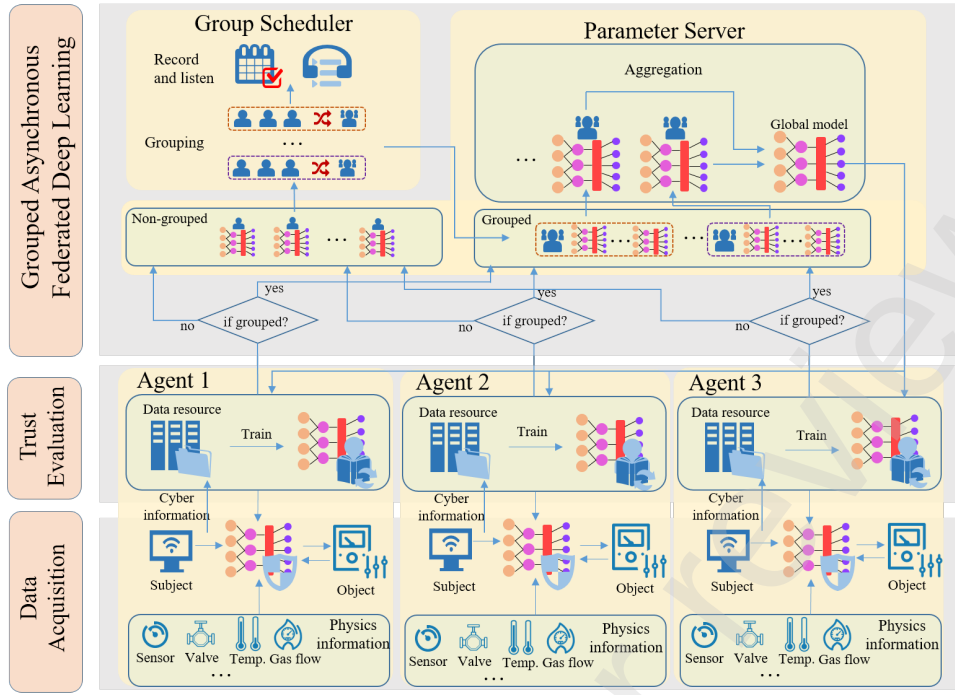


Figure 1: FedZTA system structure. *Data Acquisition:* Agents gather information from multiple dimensions and implement access control based on the decisions of the trust evaluation system, described in section 3.1. *Trust Evaluation:* The trust evaluation model based on deep learning deployed at the agents undergoes iterative training with local data, described in section 3.2. *Grouped Asynchronous Federated Deep Learning:* The trust evaluation model trained locally is aggregated with models from other agents through federated learning and dynamically updated, described in section 3.3.

trustworthiness of the subject. In addition, subject information is used to monitor and track the subject's behavior to promptly detect any suspicious activities or security risks. Subject information plays a vital role in ZTA and helps ensure the system's security and reliability.

- **Object information:** Object information refers to the information about the target resource to which the subject's requested access, marked as $\mathbf{Obj} = \{o_1, o_2, \dots, o_{no}\}$, where o is a column vector representing an attribute of the object such as the memory address and the importance. The memory address is typically used to identify the subject resource, and the importance typically refers to the influence of the resource in ICS operations and the level of privilege required to access or modify it.
- **Cyber information:** Cyber information generally refers to the traffic-related information carried by the access request itself made by the subject, e.g., IP address, network protocols, etc. Similar to subject and object information, it is marked as $\mathbf{Cyb} = \{c_1, c_2, \dots, c_{nc}\}$. Due to the connection of ICSs with the Internet, its operations and equipment are inevitably exposed to the network environment, thus potentially vulnerable to cyber-attacks. Whether it is a Denial of Service (DoS) attack that can cause system blocking or a tampering attack that can modify operational parameters,

it may cause irreparable damage to ICS. Therefore, ZTA needs to be able to identify and block cyber-attacks, and cyber information is one of the essential bases.

- **Physical information:** Physical information generally refers to the physical parameters or status of the specific ICS devices, such as temperature sensor readings, the on/off status of pressure valves, and so on. It is marked as $\mathbf{Phy} = \{p_1, p_2, \dots, p_{np}\}$. These data can reflect whether the ICS is in a stable and normal state and determine whether the equipment has suffered from attacks or abnormal operations, thus affecting the severity of the trust evaluation. That is, the physical information at the current moment can guide the trust evaluation at the next moment.

By extracting the information from the above four dimensions, the inputs to the trust evaluation model can be obtained: $\mathbf{x} = (f_1, f_2, \dots, f_n), n = ns + no + nc + np, \mathbf{x} \in \mathbf{X}, \mathbf{X} = \mathbf{Sub} \cup \mathbf{Obj} \cup \mathbf{Cyb} \cup \mathbf{Phy}$.

3.2. Dynamic and continuous trust evaluation based on deep learning model

To further enhance the performance of TES, we designed a deep learning-based policy engine. Zero trust requires a policy engine to achieve dynamic continuous evaluation and self-updating. However, the traditional policy engines of TES mainly adopt static rule matching.

Therefore, neural network models are commonly applied to achieve this goal. Considering the low latency requirements of control operations and the temporal feature of subjects, neural network models capable of extracting contextual information are needed during the training process. Moreover, when the environment (subject information, network structure, etc.) changes, the model should be able to self-update promptly while ensuring that trust evaluation operations can proceed as usual. Given these considerations, we have opted for a deep learning model combining convolutional neural network (CNN) and multilayer perceptron (MLP).

The designed trust evaluation model (referred to as the local model hereafter) mainly consists of a CNN module and an MLP module followed by a softmax layer, which is described as follows: 1) CNN Module: the CNN module mainly involves three convolutional blocks, each consisting of a convolutional layer and a maximum pooling layer. 2) MLP module: the MLP module consists of two fully connected layers and a dropout layer (used to prevent model overfitting). 3) Softmax layer: the Softmax layer maps the unnormalized output of the MLP module into a probability distribution for the prediction class.

Each industrial agent's local model utilizes the information from the four dimensions previously outlined as input to determine whether an access request can be trusted as output. If the local model requires updating, it first undergoes iterative training using local data resources.

Let j denote the training sample, $\mathbf{x}_j$ represent the features, and y_j represent the labels. The loss function is denoted as $f_j(\omega, \mathbf{x}_j, y_j)$, written as $f_j(\omega)$ for simplicity [39], where ω is the model parameter. We apply the gradient descent method to update it on dataset D_k , defined as follows:

$$F_k(\omega) = \frac{1}{|D_k|} \sum_{j \in D_k} f_j(\omega) \quad (1)$$

The learning problem is to find the optimal model ω^* that minimizes the loss function, denoted as:

$$\omega^* = \arg \min F_k(\omega) \quad (2)$$

It is almost impossible to directly solve (2), especially for deep learning models. Therefore, other methods, such as gradient descent, are usually used. The agent will perform a set of draws from the sample in each local update, update once by gradient after training, and then repeat the process to approach the optimal solution gradually. For the update of ω in the l_{th} round of local iterations, it is defined as follows:

$$\omega^l = \omega^{l-1} + \eta \nabla F_k(\omega^{l-1}) \quad (3)$$

where η is the step size of the gradient descent.

Given an access request $\mathbf{x} \in X$ from a subject as an input to the model, it views $\mathbf{x}$ as a univariate time series with multiple time steps. The processing of the CNN module is

shown in (4):

$$\begin{aligned} \mathbf{h}_1 &= \text{ConvBlock}_1(\mathbf{x}) \\ \mathbf{h}_2 &= \text{ConvBlock}_2(\mathbf{h}_1) \\ \mathbf{h}_3 &= \text{ConvBlock}_3(\mathbf{h}_2) \\ \mathbf{v}_1 &= \text{Flatten}(\mathbf{h}_3) \end{aligned} \quad (4)$$

where $\text{ConvBlock}_i, i \in \{1, 2, 3\}$ denotes the i_{th} convolutional block in the CNN module, $\mathbf{h}_1$, $\mathbf{h}_2$ and $\mathbf{h}_3$ are the hidden vectors. Then, the outputs of the three convolutional blocks are transmitted to the Flatten layer for tensorization, which results in $\mathbf{v}_1$. After the CNN module, $\mathbf{v}_1$ is fed to the MLP module, which is described as:

$$\begin{aligned} \mathbf{h}'_1 &= FC_1(\mathbf{v}_1) \\ \mathbf{h}'_2 &= FC_2(\mathbf{h}'_1) \\ \mathbf{v}_2 &= \text{Dropout}(\mathbf{h}'_2) \end{aligned} \quad (5)$$

where FC_1 and FC_2 denote the two fully connected layers and Dropout denotes the dropout layer. In addition, $\mathbf{h}'_1, \mathbf{h}'_2$ and $\mathbf{v}_2$ are the outputs of the two fully connected layers and the dropout layer, respectively. Finally, the softmax layer provides the final trust evaluation result as $\hat{y} = \text{Softmax}(\mathbf{v}_2)$. Where Softmax denotes the softmax layer and $\hat{y}$ is the final trust evaluation result of the access request.

Since the CNN model performs multi-classification to detect anomalous access requests and cyber attacks in ICSs, the cross-entropy function is used as a loss function, which is defined as:

$$\mathcal{L} = -\frac{1}{B} \sum_{j=0}^{B-1} \sum_{z=0}^{\Gamma-1} y_{j,z} \log \hat{y}_{j,z} \quad (6)$$

where B denotes the batch size, Γ denotes the number of label types, $y_{j,z}$ are the true labels, and $\hat{y}_{j,z}$ is the probability that the j_{th} example is predicted to be the z_{th} label.

3.3. Grouped asynchronous federated deep learning

Generally, the quality and quantity of training data significantly impact the performance of a model. In instances where local data resources are limited, the performance of the local model may not meet the requirements of industrial zero trust. The emerging Federated Learning (FL) technology enables data resource sharing among industrial agents while safeguarding data confidentiality. This presents a novel avenue for enhancing the performance of TES. Consequently, we extend the application of FL technology to industrial ZTA, thereby enabling TES to achieve results significantly superior to those obtained solely from local data resources.

Problem Description

Traditional synchronous federated learning (SFL) generates the optimal global model by optimizing such an

objective function [33]:

$$\min_{\omega} F(\omega) = \sum_{k=1}^m p_k F_k(\omega_k) \quad (7)$$

where F_k represents the loss function of the k_{th} agent's local model and ω_k represents its model parameter (updated by local training). p_k represents the weight factor of the k_{th} model of global aggregation, and its formula is usually set as $\frac{|D_k|}{|D|}$, $|D| = \sum_{k=1}^m |D_k|$, where $|D_k|$ represents the train data size of agent k . m represents the number of agents participating in FL. Taking FedAvg as an example, the global model ω' is formulated as:

$$\omega' = \sum_{k=1}^m p_k \omega_k \quad (8)$$

The most commonly used approach to aggregate models is simple averaging as (8) [40]. However, several drawbacks significantly negatively affect its practical application in trust evaluation. Firstly, global aggregation must be done after all agents have finished local training. However, the training speed of agents in ICS zero-trust services is inconsistent, and some agents may have an *invalid waiting delay*. It indicated that some agents need to wait for an additional period to obtain a new global model, even though they have already uploaded the model to the parameter server (PS). Furthermore, FedAvg weights global aggregation solely based on the training data size of agents. It does not sufficiently consider the influence of other variables, such as data quality, which makes it challenging to guarantee that all agents receive an equitable performance enhancement due to the aggregation process [41].

To improve aggregation efficiency and resource utilization, asynchronous federated learning (AFL) has been proposed as a promising solution [24]. It also optimizes objective (7) but eliminates the FedAvg restriction that PS must wait for all agents. Instead, each agent is allowed to aggregate with the global model individually. Specifically, the trigger condition for global aggregation changes from all agents completing local training to any one agent completing local training. When agent k uploads model ω_k to the server, the global model is updated once immediately: $\omega' \leftarrow \omega_k$. PS then returns this global model to the agent. The specific formula is as follows:

$$\omega' = (1 - \alpha_k^t) \omega^{t-1} + \alpha_k^t \omega_k \quad (9)$$

where ω' represents the global model after the t_{th} round of aggregation and α_k^t is the model update weight. In [24], α_k^t is defined as follows:

$$\alpha_k^t = \begin{cases} \alpha, & \tau_k \leq b \\ \alpha \cdot \frac{1}{a(\tau - b) + 1}, & \tau_k > b \end{cases} \quad (10)$$

where a and b are hyperparameters and α is the initial weight. τ_k is the *staleness* of ω_k , it is defined as follows:

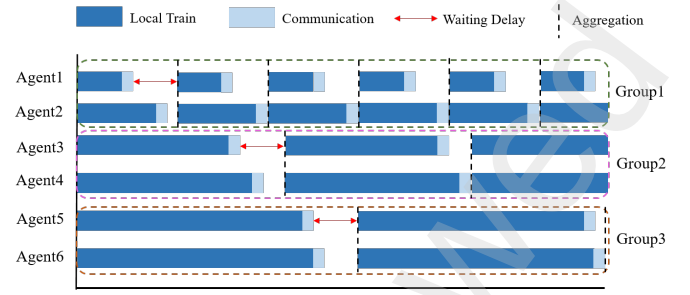


Figure 2: Illustration of the training process of FedZTA

assume that the local model ω_k is generated by the R_{kth} round of aggregation, while the current aggregation round is t , we refer to $\tau_k = t - R_k$ as the staleness of ω_k . (10) shows that the model weights decrease rapidly with increasing staleness when $\tau_k > a$.

This completely asynchronous approach results in a minimal invalid waiting delay of agents, significantly improving aggregation efficiency and saving time overhead. However, unconditional asynchronous aggregation has several non-negligible drawbacks in ICS zero trust as follows: Firstly, agents that train faster will get more opportunities for aggregation, thus making the global model overly biased towards them, resulting in poorer model performance. Secondly, this approach results in a much higher frequency of global aggregation than in FedAvg, leading to a significant increase in communication overhead.

Structure and process of the FedZTA

In response to the problem outlined in 3.3, we develop a grouped asynchronous federated learning model for ICS zero-trust architecture (FedZTA). It is implemented by two main components: the scheduler and the server. On the other hand, the scheduler is responsible for listening to the server's cache, dividing the ungrouped models in it, and controlling the aggregation of the grouped models. The server is responsible for receiving local models from different industrial agents and aggregating them according to the scheduler's commands. The specific grouping and aggregation process is illustrated in Fig.2.

A fundamental principle to be adhered to in the process of FL grouping is that agents with similar *response latencies* (i.e., the length of time elapsed by an agent from the time it receives the global model to the time it finishes local training and uploads the model to PS) should be classified into a single group as far as possible. However, the response latency is not known a priori, which presents a challenge for grouping. The treatment in most current studies can be categorized into two approaches: 1) Before FL, separate tests are conducted to collect the response latency from each agent. This approach ensures excellent reliability as it utilizes actual latency. However, it entails additional time and communication overheads. 2) The response delay is calculated in advance using specific attributes (e.g., agent arithmetic, data size, communication distance, etc.). This

approach necessitates minimal additional time overhead, yet the calculated data may be subject to error due to the multitude of factors influencing response latency. Furthermore, agents must convey local device attributes and other data to PS, increasing the risk of privacy leakage.

In light of the preceding analysis, a novel grouping strategy is proposed. It does not require a separate delay collection or the agents' attribute information to be collected in advance. Instead, it integrates the grouping algorithm with the entire FL process, thereby addressing the two issues mentioned above.

Our grouping algorithm is mainly implemented through a group scheduler (GS). It is used to monitor the grouping status of agents during FL. Initially, all agents' group indexes C_k are set to 0, meaning they are non-grouped. GS records the current number of groups $g = 0$. When GS listens for the first model with $C_k = 0$ to reach the server buffer (the agent is denoted as A), it opens a waiting delay of length T . During T time, all other agents with $C_k = 0$ uploaded to PS are divided into the same group as agent A . When T runs out, GS sets $g = g + 1$, $i = g$ and records the number of models with $C_k = i$ in the buffer at this time (i.e., records N_i , the number of agents in the group i), and the maximum response latency of the group i : $L_i = L_A + T$, where L_A is the response latency of agent A .

After that, GS continuously listens to the response status of each group and controls the subsequent aggregation conditions. GS sends an aggregation command to PS when the grouped agents (i.e., $C_k \neq 0$) satisfy any one of the following two conditions: 1) All N_i agents in the group have arrived at the PS buffer. 2) The response latency exceeds L_i . This way of controlling PS aggregation via the scheduler has the following two advantages: Firstly, grouping based on the maximum waiting delay T ensures that the invalid waiting delay of the agents in each group will not exceed T . This delay can be further minimized in subsequent rounds of FL, i.e., the aggregation is executed when all the agents in the group have completed the training (satisfying Condition 1). Secondly, the maximum response latency L_i in Condition 2 can enhance the robustness of FL. When an agent drops out (i.e., the agent cannot complete the local training and the subsequent FL process), it will not affect the other agents in the group to continue participating in FL.

In our FedZTA model, PS is mainly responsible for collecting and aggregating the deep learning model uploaded by each industrial agent and returning the new global model to the corresponding agent. PS has a parameter buffer that stores the following key attributes about agents:

- k : agents index
- ω_k : local model of agent k ,
- R_k : the aggregation round when the agent last received the global model,
- C_k : the group to which agent k belongs,
- t : current aggregation round,

- ω^t : current global model.

When GS sends an aggregation command to inform PS of the group that needs to be aggregated, and then PS performs one intra-group aggregation for the models in that group as (11):

$$\omega_i = \sum_k \frac{N_i}{\sum_k N_i} \omega_k \quad (11)$$

where d_k denotes the degree of change in the model, and it is defined as $d_k = \frac{1}{\eta}(\omega_k^0 - \omega_k)$, ω_k^0 represents the global model received by agent k without local training. Since groups follow the principle of synchronous aggregation, the effects of staleness do not have to be considered between the models used by agents in the same group. Intra-group aggregation can reduce the number of stale models and the model bias, which makes the models more valuable in the global aggregation.

Then, PS performs global aggregation according to (12) immediately after intra-group aggregation:

$$\omega^t = (1 - \alpha_i^t) \omega^{t-1} + \alpha_i^t \omega_i \quad (12)$$

where the weighting factor α_i^t is determined by (13):

$$\alpha_i^t = (1 - \frac{g-1}{m}) \tau_i^{-1} \rho_i \quad (13)$$

where ρ_i is a weight related to the agent's training speed, which is defined as follows:

$$\rho_i = 1 - \frac{\frac{E}{L_i}}{\sum_{j=1}^g \frac{E}{L_j}} \quad (14)$$

where E is the overall time budget. ρ_i is such that groups with faster training will receive smaller weights in the aggregation, preventing the global model from being overly biased by too many aggregations of that group. This approach causes the model weight of each group to decrease as the number of groups increases. It is obvious that when all participating agents belong to the same group, $g = 1$, which is equivalent to FedAvg, when all participating agents do not belong to the same group, $g = m$, which is comparable to FedAsync.

After that, the current aggregation round is updated: $t = t + 1$, and the global model is updated to the result of (12). Then, PS packages C_k , t , ω^t and sends them to the agents participating in this aggregation. Algorithm 1 shows the FL process for FedZTA containing grouping and aggregation.

Undoubtedly, in scenarios where there are differences in the response latency of agents, our algorithm significantly reduces the time expenditure required to complete the specified number of global aggregations compared to SFL. Assuming the need to perform h global aggregations, with a maximum response latency of agents as L_{max} , and the communication time cost between agents and servers

along with the server computation time cost as $T(c_1 + c_2)$, the total time cost of SFL is $h(L_{max} + T(c_1 + c_2))$. Given that $T(c_1 + c_2) \ll L_{max}$, it can be stated that the time complexity of SFL is $O(hL_{max})$. As for FedZTA, the time complexity is $O(\sum_i^g h_i L_i)$, where h_i represents aggregation rounds participated by the i_{th} group, and $\sum_i^g h_i = h$. Assuming that in h global aggregations, the group with the maximum response latency can participate in at least h' aggregations, it can be deduced that $h_i = \frac{L_{max}}{L_i} h'$. So the time complexity of FedZTA can be expressed as $O(h' L_{max})$, $h' < h$.

Algorithm 1 Local models update based on FL.

INPUT: T : maximum waiting delay during group partitioning, m : number of agents, h : number of global aggregations, E : time budget
 OUTPUT: ω^t : global model

initialize the number of groups $g = 0$, current round $t = 0$

```

while  $t < h$  :
  agents train local model  $\omega_k$ 
  PS receives  $\omega_k$ 
  if  $C_k = 0$ :
    set  $g = g + 1, i = g$ 
    record current time  $cT_1$ 
    listening to the server buffer
    if current time  $cT_2 - cT_1 \geq T$ :
      select all models with  $C_k = 0$ 
      change  $C_k = i$ 
       $L_i = L_A + T$ 
      record numbers of agents  $N_i$ 
      send aggregation command to PS:  $C = i$ 
    if  $N'_i == N_i$  or  $cT - cT_i \geq L_i$  ( $cT$  means current time):
      GS send  $C = i$  to PS
    if PS receives  $C = i$  from GS:
       $\omega_i \leftarrow \sum_k^{N_i} \frac{d_k}{\sum_k^{N_i} d_k} \omega_k$ 
       $\tau_i \leftarrow t - R_k$ 
       $\rho_i \leftarrow 1 - \frac{\frac{E}{L_i}}{\sum_{j=1}^g \frac{E}{L_j}}$ 
       $\omega^t \leftarrow (1 - \alpha_i^t) \omega^{t-1} + \alpha_i^t \omega_i$ 
      listening to the server buffer
      send  $\omega^t$  to the agents in group  $i$ 
       $cT_i = cT$ 
  end while
return  $\omega^t$ 

```

Convergence Analysis

To analyze the convergence of FedZTA, we first make some widely used assumptions as follows:

Assumption 1. Assume that the loss function F is μ - strongly convex, that is, for any independent variables $\omega_1, \omega_2 \in \mathcal{R}^d$, the following equation holds:

$$F(\omega_1) - F(\omega_2) \geq \nabla F(\omega_2)^T + \frac{\mu}{2} \|\omega_1 - \omega_2\|^2$$

Assumption 2. Assume that the loss function F is L - smooth, that is, for any independent variables $\omega_1, \omega_2 \in \mathcal{R}^d$, the following equation holds:

$$F(\omega_1) - F(\omega_2) \leq \nabla F(\omega_2)^T + \frac{L}{2} \|\omega_1 - \omega_2\|^2$$

Next, we analyze the convergence of the global model during local iterative training.

Theorem 1. When loss function F satisfied $\mu \leq \frac{1}{L}$, after l iterations of local training, the convergence bound is:

$$F(\omega^l) - F(\omega^*) \leq (1 - \eta\mu)^l (F(\omega^0) - F(\omega^*))$$

Proof. First analyze the single-step iterative training, from (3) and Assumption 2 it follows that:

$$\begin{aligned}
 F(\omega^l) - F(\omega^{l-1}) &= F(\omega^{s-1} + \eta \nabla f(\omega^{l-1})) - F(\omega^{l-1}) \\
 &\leq \nabla F(\omega^{l-1})^T (\eta \nabla F(\omega^{l-1})) + \frac{L}{2} \|\eta \nabla F(\omega^{l-1})\|^2 \quad (15) \\
 &\leq \left(\frac{L}{2} \eta^2 - \eta\right) \|\nabla F(\omega^{l-1})\|^2
 \end{aligned}$$

From Assumption 1, we can derive the following:

$$2\mu(F(\omega_1) - F(\omega_2)) \leq \|\nabla F(\omega_1)\|^2 \quad (16)$$

Combining (15) and (16), (17) can be derived:

$$\begin{aligned}
 F(\omega^l) - F(\omega^{l-1}) &\leq \eta(1 - \frac{L}{2}) 2\mu(F(\omega^{l-1}) - F(\omega^*)) \quad (17) \\
 &\leq -\eta\mu(F(\omega^{l-1}) - F(\omega^*))
 \end{aligned}$$

Then, (18) is derived by plus an $F(\omega^*)$ to the left and right sides:

$$F(\omega^l) - F(\omega^*) \leq (1 - \eta\mu)(F(\omega^{s-1}) - F(\omega^*)) \quad (18)$$

Finally, the conclusion of Theorem 1 can be introduced by mathematical induction, and the proof is complete.

We then analyze the convergence of the model aggregation phase.

Theorem 2. When loss function F satisfied $\mu \leq \frac{1}{L}$, after l iterations of local training and t global aggregations, the convergence bound is:

$$F(\omega^l) - F(\omega^*) \leq ((1 - \frac{g-1}{m})(1 - \eta\mu)^l + \frac{g-1}{m}) (F(\omega^0) - F(\omega^*))$$

Proof. Based on (11) and (12), the following conclusion can be introduced:

$$\begin{aligned}
 F(\omega^t) - F(\omega^*) &\leq (1 - \alpha_i^t) F(\omega^{t-1}) + \alpha_i^t F(\omega_i) - F(\omega^*) \\
 &\leq (1 - \alpha_i^t) F(\omega^{t-1}) + \alpha_i^t F\left(\sum_k^{N_i} \frac{d_k}{\sum_k^{N_i} d_k} \omega_k\right) - F(\omega^*) \quad (19) \\
 &\leq (1 - \alpha_i^t) F(\omega^{t-1}) + \alpha_i^t F(\omega_k) - F(\omega^*) \\
 &\leq F(\omega^{t-1}) - F(\omega^*) + \alpha_i^t (F(\omega_k) - F(\omega^{t-1}))
 \end{aligned}$$

From the second and fifth terms, we note that the loss of the aggregated model of each group i is smaller than that of any agent k in this group. This is because the aggregated model learns information from all local datasets of agents in group i while the local model of agent k is only trained over the local dataset.

We assume that the global model used by agent k for local training at the time of the current round of aggregation is the global model from the previous round ω^{t-1} , it is clear that ω^{t-1} is equivalent to ω^0 in Theorem 1, and ω_k after local training is equivalent to ω^l in Theorem 1. Then, the following conclusions can be drawn:

$$\begin{aligned} F(\omega^l) - F(\omega^*) &\leq F(\omega^{t-1}) - F(\omega^*) + \alpha_i^t((1 - \eta\mu)^l - 1)(F(\omega^{t-1}) - F(\omega^*)) \\ &\leq (\alpha_i^t(1 - \eta\mu)^l - \alpha_i^t + 1)(F(\omega^{t-1}) - F(\omega^*)) \end{aligned} \quad (20)$$

From (13), we know that $\alpha_i^t \leq (1 - \frac{g-1}{m})$. By the mathematical method of induction, the following conclusions can be drawn:

$$\begin{aligned} F(\omega^l) - F(\omega^*) &\leq ((1 - \frac{g-1}{m})(1 - \eta\mu)^l + \frac{g-1}{m})(F(\omega^{t-1}) - F(\omega^*)) \quad (21) \\ &\leq ((1 - \frac{g-1}{m})(1 - \eta\mu)^l + \frac{g-1}{m})^t(F(\omega^0) - F(\omega^*)) \end{aligned}$$

3.4. The overall process of FedZTA

In the FedZTA system, agents first deploy local models (derived from the global model obtained in the previous FL participation) and then iteratively train the model with local data resources. The relevant information is extracted from multiple dimensions and used as input $\mathbf{x}$ for the local model when handling subject access requests. Subsequently, the local model assesses the trustworthiness of the access request, which is a continuous process in ZTA operations. Additionally, agents can continue to participate in FL to dynamically update their local models to adapt to environmental changes.

4. Experimental and Analysis

4.1. Experimental environment

In the work of this paper, the designed CNN-based policy engine model is implemented using Pytorch, and the federated learning framework is built by the lightweight Python framework FLgo. Our experiments were conducted on the Ubuntu 18.04.3 LTS platform with an Intel(R) Core(TM) i7-10700 CPU and an NVIDIA GeForce GT730 GPU (8 GB RAM).

4.2. Resource description

GPN dataset

We use the dataset on a real gas pipeline industrial system suitable for ICS zero-trust operations, named the GPN dataset.

Algorithm 2 The overall process of FedZTA.

INPUT: D_k : train data, $\mathbf{X}$: set of access requests, l : local train epochs

OUTPUT: $\hat{y}$: trust evaluation result

(Dynamic updating of local model)

Algorithm 1 generates global model ω^t

agents receive global model ω^t from parameter server

set $\omega_k = \omega^t, u = 0$.

while $u < l$:

calculate loss $F_k(\omega_k) \leftarrow \frac{1}{|D_k|} \sum_{j \in D_k} f_j(\omega_k)$

$\omega_k \leftarrow \omega_k + \eta \nabla F_k(\omega_k)$

$u = u + 1$

end while

(Continuous trust evaluation)

for $\mathbf{x}$ in $\mathbf{X}$:

$\mathbf{v}_1 \leftarrow CNNmodule(\mathbf{x})$ (as Eq.(4))

$\mathbf{v}_2 \leftarrow MLPmodule(\mathbf{v}_1)$ (as Eq.(5))

$\hat{y} \leftarrow Softmax(\mathbf{v}_2)$

if $\hat{y} == 0$:

access request $\mathbf{x}$ is trusted

else:

access request $\mathbf{x}$ is untrusted

end for

The gas pipeline system aims to transmit natural gas from a gate station to a residential dwelling. Its physical environment consists of a gas pump that mimics a gate station, a section of pressure piping, three pressure converters, three pressure shut-off valves, and multiple sensors for monitoring pressure, flow rate, and temperature. Among them, a gas pump provides high-pressure gas for the physical environment; gas piping is divided into three sections, namely high-pressure piping, medium-pressure piping, and low-pressure piping, each of which is controlled by a pair of pressure transducers and switches to control the drop in gas pressure; multiple sensors are responsible for collecting observations such as temperature, flow rate and pressure in the piping; and a safety instrumented security system protects the functional safety of the system. In addition, the scenario utilizes two PLCs and an operator control station to control the entire legal business process. The business process is as follows: turn on the gas pumps to generate high pressure; open all valves in sequence to build a gas pipeline delivery path from the gate station to the residents' homes; and turn on all pressure converters to reduce the gas pressure to a usable range for the residents. The specific process is shown in Fig.3.

Data features consist of four parts: subject information, object information, cyber information, and ICS physical information. The composition of the features corresponds to the multi-dimensional information presented in Section 3.1, the details of which are shown in A.

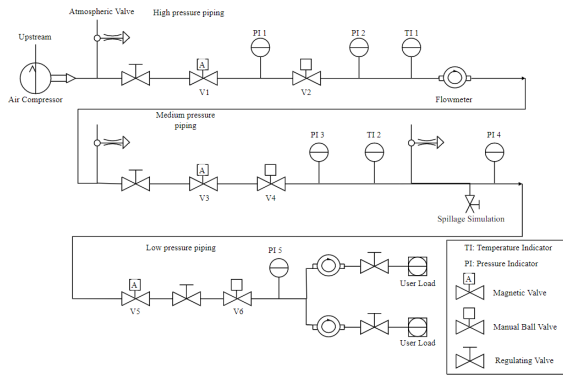


Figure 3: GPN testbed structure

WADI dataset

WADI is a vital industrial control system (ICS) security research dataset. It simulates an industrial control system for water supply distribution and records various operations and events in the system. This dataset mainly evaluates and develops security solutions for industrial control systems and studies possible attacks and vulnerabilities in ICSs.

The WADI testbed is equipped with a chemical dosing system, booster pumps and valves, instrumentation, and analyzers [42]. In addition to simulating attacks and defenses against PLCs over the network, WADI can also simulate the effects of physical attacks, such as water leaks and malicious chemical injections. The WADI data collection process consisted of 16 days of continuous operation, 14 days of which were collected under regular operation and two days under attack. All network traffic, sensor, and actuator data is collected during the data collection period. For more detailed information on the WADI dataset, see the WADI website.

4.3. Data partitioning and response latency simulation

In the experiments, the data resources are divided into two main parts, i.e., 80% for training and 20% for testing, and the training part is further divided non-uniformly to each industrial agent for local model training. Note that all trained deep learning models are tested on the same test data and the data distribution remains consistent when performing experiments with different comparison algorithms.

We set up a total of 20 industrial agents, and due to the limited number of devices, multiple agents need to run on the same machine. This leads to the fact that we cannot use training times to represent actual training times in real business environments because they share computational resources. To solve this problem, we set a fixed response latency for each industrial agent, which is used to simulate the training time and transmission time of an industrial agent in a real business. We stipulate that the response latency of 20 industrial agents is uniformly distributed within 5s-20s, i.e., the fastest-trained agent responds with a latency of 5s. In comparison, the slowest trained agent responds with a

Table 2
Summary of Main Notations

Algorithm	Symbol	Description	Value
FedZTA	T	Maximum waiting delay	3s
FedAsync[24]	α	Initial weight	0.6
	a	Staleness factor	10
	b	Stale limit	4
Fedbuff[25]	K	Number of models in the buffer	5
TiFL[21]	T_{ier}	Number of tiers	5
FedMgda+[34]	I	Intervals for changing probability	10s
	η_1	Global step size	0.5
	ϵ	Attenuation factor	0.1
SCAFFORD[22]	η_2	Global step size	0.1

latency of 20s. In practice, there are a lot of factors affecting the training speed of the agents, such as the amount of data, the computational power of the device, etc., but they are all reflected in the response latency in the end, so we can directly set this element to perform simulation of the real environment.

4.4. Parameters and Evaluation metrics

In this work, we compare the performance of our proposed FedZTA scheme with some state-of-the-art studies and also with classical synchronous federated learning (FedAvg) and asynchronous federated learning (FedAsync) from several perspectives.

The comparison algorithms all use the same policy engine model as FedZTA, which has a learning rate lr set to 0.1. Regarding their respective unique hyperparameters, we refer to the corresponding literature as well as analyze them experimentally to arrive at the settings that perform best in our experimental setting. The following table lists the unique hyperparameter settings of the compared algorithms.

The five main evaluation indicators used in this experiment are as follows:

- **Accuracy:** The accuracy of the global model for classification on the test set and is used to evaluate the overall classification performance of the model.
- **Time cost:** Time spent on the entire federal learning process.
- **Variance:** The variance of the global model's local test accuracy across agents, which is used to evaluate the stability of the model and the fairness of the FL algorithm.
- **Test Time:** The computational time consumed by the global model to test it on the test set, and it is used to evaluate the speed of decision-making by the policy engine based on this model.

4.5. Result analysis

On GPN dataset

We first compared the performance of the seven algorithms after 50 global aggregations on our GPN dataset with

Table 3

Results of 50 aggregations on the GPN dataset

Type	Algorithm	Accuracy	Time cost	Variance	Test time
SFL	FedAvg	0.9250	1003	0.0242	0.2175
	TiFL	0.9216	544	0.0262	0.2239
	FedMgda+	0.9089	1003	0.0207	0.1791
	SCAFFORD	0.9243	1003	0.0237	0.2324
AFL	FedAsync	0.8721	38	0.0282	0.2219
	Fedbuff	0.9134	118	0.0285	0.2214
	FedZTA	0.9384	132	0.0156	0.2244

all features. The epoch of local training of the models is 20. The results are shown in Table 3.

For comparison, SFL and AFL are divided into two separate sections in the table. It can be observed that when there are differences in agent response delays, the time cost required for AFL algorithms to complete the specified aggregation rounds is much lower than that of SFL algorithms. Among them, the FedAsync with the lowest time cost takes only 38 seconds to complete 50 aggregations, but at the expense of accuracy, achieving only 0.8721, and it's an unstable value. Fedbuff, which utilizes a cache waiting mechanism, improves accuracy at the cost of a slightly higher time cost, achieving an accuracy of 0.9134 with a time cost of 118 seconds compared to FedAsync. Our FedZTA algorithm not only further improves accuracy to 0.9384 but also controls the time cost to a reasonable 132 seconds. Additionally, due to FedZTA's consideration of bias in the asynchronous aggregation process, enhances the fairness of the global model, as evidenced by the lowest variance value of 0.0156. Compared to AFL, SFL generally exhibits higher overall accuracy but at a significantly higher time cost. Even TiFL, which uses hierarchical sampling to expedite the FL process, incurs a time cost of 544 seconds. It is worth noting that, except for FedMgda+, the testing time for all algorithms is around 0.21-0.23. Given that the test set contains 4600 samples, this testing time indicates that the trust evaluation system based on CNN-MLP can make judgments on an access request in a very short time, meeting the real-time requirements of ICS services.

It should be noted that the GPN dataset used in this data set was collected in a real industrial environment. Due to the stability of the experimental system, anomalous access was not permitted to affect the real industrial system, and thus the collected physical domain information is representative of normal values. Furthermore, simulation experiments were conducted to observe the effect of physical domain characteristics on the trust evaluation system. In these experiments, the physical domain characteristics of the anomaly access were modified to a random distribution within an anomaly range. The results of the simulation experiments are presented in Table 4.

From observation, it can be seen that the inclusion of physical information features has led to an improvement in the accuracy of all algorithms. Four algorithms have

Table 4

Results of 50 aggregations on the simulated GPN dataset

Type	Algorithm	Accuracy	Time cost	Variance	Test time
SFL	FedAvg	0.9514	1003	0.0225	0.2195
	TiFL	0.9531	544	0.0244	0.2288
	FedMgda+	0.9231	1003	0.0199	0.1811
	SCAFFORD	0.9541	1003	0.0253	0.2417
AFL	FedAsync	0.8916	38	0.0238	0.2231
	Fedbuff	0.9377	118	0.0217	0.2207
	FedZTA	0.9547	132	0.0162	0.2312

achieved accuracy levels above 0.95 (FedAvg, TiFL, SCAFFORD and FedZTA with accuracy levels of 0.9514, 0.9531, 0.9541 and 0.9547 respectively). Of these, only FedZTA belongs to AFL and has a lower time cost. It is worth noting that the simulated GPN experiments, compared to the real GPN experiments, only changed the values of the physical features without adding new features, resulting in minimal changes in variance and testing time. As the time cost was simulated under the same response delay conditions, it remained unchanged.

Due to the huge difference in the time cost required for SFL and AFL to aggregate the specified rounds under our heterogeneous simulation conditions, just unifying the aggregation rounds for experimentation is not comprehensive. So, we unified time budget $E = 200s$ and conducted a more detailed and multi-faceted comparative experiment, to analyze convergence speed and accuracy performance between FedAvg and FedAsync, two representative FL algorithms used as benchmarks, and FedZTA. The results are shown in Fig.4.

According to Fig.4(a), it is evident that FedZTA exhibits a significant advantage in convergence speed, reaching an accuracy of over 0.90 by 40 seconds. The time taken for all three algorithms to reach a relatively stable accuracy is around 60 seconds. This is due to FedZTA's grouping mechanism balances data resource utilization and time cost. The accuracy after stabilization is relatively close, with FedZTA achieving the highest accuracy of 0.9621. Within the first 80 seconds, both FedZTA and FedAsync outperform FedAvg in accuracy, attributed to the aggregation mechanism of AFL allowing them to aggregate more times in a relatively shorter period. Ultimately, FedAsync has the lowest accuracy, attributed to the model bias issue resulting from incomplete asynchronous handling.

Fig.4(b) shows the results obtained after removing the subject and object information features from the original training data. In comparison to Fig.4(a), it can reflect the impact of subject and object information on federated ZTA. The results indicate a significant overall decrease in accuracy, with the highest being FedZTA at only 0.8601. This suggests that subject and object information are crucial factors for trust evaluation in ZTA access requests. It also demonstrates the feasibility of using these as sources of information input to neural network models for training. It is worth noting that FedAsync exhibits significant accuracy

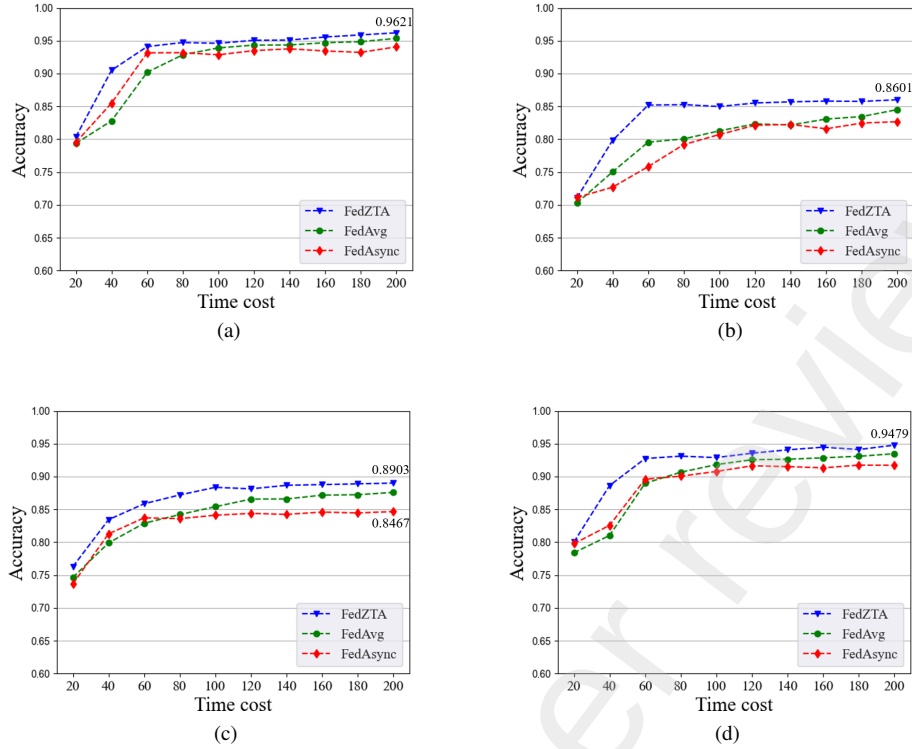


Figure 4: Figure 4(a) illustrates the global model accuracy over time for the three algorithms using all multi-dimensional information. Figure 4(b), 4(c), 4(d) illustrate the three scenarios with no subject and object information, no network information and no physical information, respectively.

fluctuations at 200s of testing, while FedZTA and FedAvg remain relatively stable, indicating a deeper level of utilization of other information by FedZTA and FedAvg.

Fig.4(c) presents the results obtained after removing cyber information features from the original training data. In comparison to Fig.4(a), it can reflect the impact of cyber information on federated ZTA. Since our experiments simulated a cyber-attack environment, cyber information is crucial for identifying these attacks. As expected, upon removing cyber information, the accuracy of all algorithms decreased, with the highest being FedZTA at 0.8903. Furthermore, both FedZTA and FedAvg showed a noticeable decrease in convergence speed, with FedZTA stabilizing after 100 seconds and FedAvg after 120 seconds. Although FedAsync was able to maintain stability after 60 seconds, its accuracy significantly declined, with a maximum value of only 0.8467. It can be confirmed that cyber information is one of the essential information sources for training zero-trust models in a cyber environment.

We similarly validated the importance of physical information, with the results shown in Fig.4(d). The overall convergence trends of the three algorithms were similar to those in Fig.4(a), but the accuracy decreased. This indicates that physical information can indeed enhance the performance of ICS trust evaluation systems. Overall, FedZTA achieved the highest accuracy and the fastest convergence speed in all scenarios, demonstrating its better adaptation

to the business requirements of ICS zero-trust compared to FedAvg or FedAsync. This also confirms that the fusion of multi-dimensional information is an effective means to enhance the performance of trust evaluation models.

In the self-optimization of the FedZTA algorithm, we conducted a comparative analysis from two perspectives. We analyzed the impact of changing the maximum waiting delay T by examining the changes in accuracy and time cost over 50 aggregation rounds of FedZTA. We considered three different choices for $T \in \{2, 3, 4\}$, as depicted in Fig.5. Observing from Fig.5(a), it is evident that altering T has not only an effect on the convergence speed of FedZTA but also has a certain impact on accuracy. This is because increasing T increases the number of agents within each group, thereby utilizing more data resources, leading to an improvement in accuracy and a reduction in aggregation rounds. However, this improvement is not limitless; the accuracy at $T = 3$ is 0.9502, significantly better than the accuracy at $T = 2$, which is 0.9359, while the accuracy at $T = 4$ is 0.9537, showing less pronounced superiority over $T = 3$. Fig.5(b) analyzes the variation in time cost for different T , indicating that as T increases, the time cost also rises. This is because expanding T leads to an increase in the ineffective waiting delay of some agents within the group, as described in detail in Section III, confirming the validity of our theoretical framework.

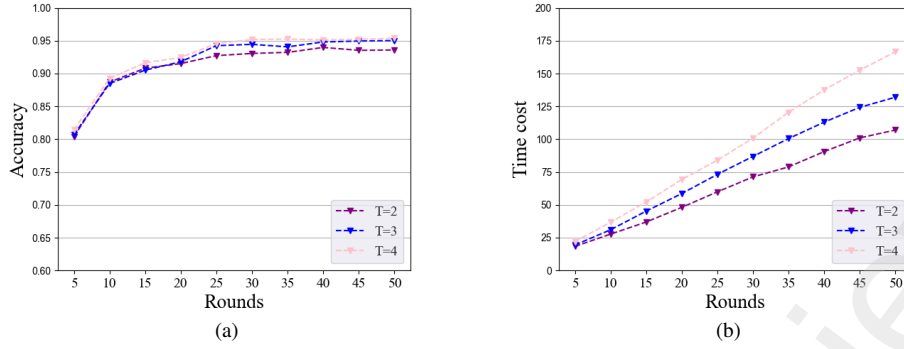


Figure 5: Results of accuracy and time cost for different T

Table 5
Results of 50 aggregations on the WADI dataset

Type	Algorithm	Accuracy	Time cost	Variance	Test time
SFL	FedAvg	0.9959	802	0.0069	0.4012
	TiFL	0.9946	401	0.0076	0.4121
	FedMgda+	0.9876	802	0.0061	0.3017
	SCAFFORD	0.9962	802	0.0082	0.4543
AFL	FedAsync	0.9301	28	0.0157	0.3731
	Fedbuff	0.9711	101	0.0071	0.3981
	FedZTA	0.9801	107	0.0068	0.3876

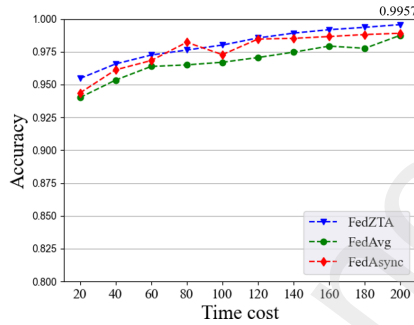


Figure 6: Accuracy results over 200s on the WADI dataset

On WADI dataset

In this experiment, we used the data collected under attack and employed the sliding window sampling method proposed in [43] to extract 20% of the data for the experiments, and utilised all 127 features in the original dataset, which encompasses both cyber features and physical domain features. The results are shown in Table 5.

Due to WADI being an intrusion detection dataset in the ICS field rather than specifically tailored for zero-trust business, its data characteristics lack fundamental subject and object information crucial for trust evaluation tasks, containing only network and physical information. Moreover, with a plethora of 127 features, achieving stable convergence in accuracy for AFL requires significantly more aggregation rounds. Under the condition of 50 aggregations, SFL generally outperforms AFL in accuracy, with the highest

SCAFFORD reaching 0.9962, while the lowest FedAsync is at 0.9301. However, AFL's advantage lies in its accelerated aggregation efficiency, with considerably lower time costs compared to SFL. In AFL, FedZTA attains an accuracy of 0.9801, surpassing FedAsync and Fedbuff.

As analyzed in the preceding section, AFL's strength lies in generating more aggregations within a specified time budget. Hence, we conducted a comparative analysis under the given time budget $E=200s$, as illustrated in Fig.6. It is evident that when transitioning to a fixed-time budget scenario, AFL's advantage becomes apparent. Overall, the performance results indicate FedZTA > FedAsync > FedAvg, with FedZTA exhibiting a relatively smooth convergence trend, while FedAsync shows comparatively unstable behavior. At 200 seconds, FedZTA achieves an accuracy of 0.9957, matching the SFL level in Table 5, yet with significantly lower time costs. It demonstrates that FedZTA can achieve the desired effects on the WADI dataset by substantially reducing FL time costs while maintaining accuracy levels.

5. Conclusion

This paper proposes a zero-trust architecture based on federated deep learning for access control in industrial control systems, named FedZTA. In this architecture, industrial agents deploy a deep-learning neural network model as the policy engine for the trust evaluation system. Multiple sources of information are extracted from the operational environment as inputs to the local model and used to continuously evaluate the trust of access requests. Additionally, the agents' local models achieve dynamic updates through a novel grouped asynchronous federated learning approach. The effectiveness of FedZTA is validated in a real gas pipeline system, demonstrating faster convergence speed and higher accuracy compared to existing FL-based ZTA approaches. Future work may involve designing a mechanism for adaptive trade-offs between the convergence speed and accuracy of FedZTA to address varying ICS business requirements.

Table 6
Subject information

Feature	Description	Purpose
User Id	Entity that issued the access request	User Id is used to determine the identity of the accessing entity and its role in the system.
Role	A collection of all roles created in the system	Roles simplify ICS management and enhance security, satisfying the principle of segregation of duties.
Location	The subject's geographic location.	For example, specific tasks can only be performed by subjects within certain designated areas.
Time	User access time.	For example, some industrial control equipment may have restricted opening hours, thereby reducing the likelihood of risk, workload and energy consumption.
OsVer	Operating system version of the subject	Keeping abreast of the operating system versions of industrial control devices ensures timely response to known vulnerabilities and enables more accurate authorization assessment.
SwVer	The software version of the subject.	The software version relates to the safety and reliability of the subject's operations.
Geo Cred	Geolocation credibility.	Improve the accuracy of authorisation assessments based on whether a geographic location is under threat.
Frequency	Frequency of access	Monitoring the frequency of accesses helps to detect anomalous behaviour and enhances the security of the model.
Delay	The time spent during the subject's access.	Helps to optimise production planning, reduce the risk of network congestion and optimise resource utilisation.

Table 7
Object information.

Feature	Description	Purpose
Address	Memory address of the objects	Maps objects to PLC memory addresses for easy management and access.
Importance	Importance of the objects	Classify objects according to their sensitivity and safety.

A. Detail description of the GPN dataset

Features

The data features consist of four parts: subjects information, objects information, cyber information and physical information, shown in table 6, table 7, table 8, table 9, respectively.

Data Settings

Set up 50 User Id, there are 45 normal users: user 1 to user 45 and 5 abnormal users: abnormal user 1 to abnormal user 5. Object resources: gas flow, temperature, high pressure, medium pressure, low pressure, low pressure cut-off valves, medium pressure cut-off valves, high pressure cut-off valves as the access object. The roles are divided into Temp role, Role1-Role3, and the correspondence between roles and objects and users is shown in Table6. User access locations are selected as WorkPlace, Home, StrangePlace1-10: where StrangePlace 1 and StrangePlace 2 are contaminated.

Data collection was performed in three scenarios: normal user access in normal scenario, abnormal user access in normal scenario and normal user access in three attack scenarios. The ratio of normal user access and abnormal user access in normal scenarios is six to four, totalling 20,000 entries. The normal user access in the attack scenario is 3,000 pieces of data, including three common types of ICS cyber-attacks.

- Normal user access in normal scenario: Random User1-45, random subject info in authority, Location(95% WorkPlace, 4.9% Home, 0.1% StrangePlace), Time(95% 8-20h, 4% 7-8,20-22h, 1% else), OsVer(95% new, 5% old), SwVer(95% new, 5% old).
- Abnormal user access in normal scenario: Random abnormal user 1 - abnormal user 5, Random subject info, Location(4% WorkPlace, 1% Home, 95% StrangePlace), Time(%1 8-20h, %4 7-8,20-22h, %95 else), OsVer(%5 new, %95 old), SwVer(5% new, 95% old).
- Normal user access in three attack scenarios:
 - Dos attack. Random User1-45 Random guest in authority Location(95% WorkPlace, 4.9% Home, 0.1% StrangePlace), Time(95% 8-20h, 4% 7-8,20-22h, 1% else), OsVer(95% new, 5% old).
 - Theft Attack. Random User1-45 Random guest in authority Location(95% WorkPlace, 4.9% Home, 0.1% StrangePlace), Time(95% 8-20h, 4% 7-8,20-22h, 1% else), OsVer(95% new, 5% old).
 - Fake data injection attack where a random physical information is chosen to be changed. Random User1-45 Random guest in authority

Table 8
Cyber information

Feature	Description
frame.time.utc	Packet capture time
ip.src	Source IP address
tcp.srcport	Source TCP port
ip.dst	Destination IP address
tcp.dstport	Destination TCP port
frame.protocols	Protocol stack information for data packets
frame.time.delta	The time difference between the data packet and the previous data packet
frame.time.relative	The time difference between the data packet and the start time of crawling
frame.time.delta.displayed	The time difference between the packet and the previous packet (displayed)
frame.len	packet length
eth.src.oui	Manufacturer identification code for the source MAC address
eth.dst.oui	Manufacturer identification code for the destination MAC address
tcp.len	packet length
tcp.ack	Acknowledge number of TCP packets
tcp.analysis.bytes.in.flight	The number of unconfirmed bytes on the network
tcp.analysis.ack.rtt	Measure the captured TCP packets and their corresponding ACKs

Table 9
Physical information

Feature	Description
Gas flow rate	The amount of gas flowing through a pipe at a given time
Temperature	The temperature status of the pipes
Medium Pressure	Pressure state of a medium pressure pipe
High Pressure	Pressure state of a high pressure pipe
Medium pressure shut-off valve	Status of medium pressure valves for control of medium pressure pipelines
High pressure shut-off valve	Status of high pressure valves for control of high pressure pipelines

Location(95% WorkPlace, 4.9% Home, 0.1% StrangePlace), Time(95% 8-20h, 4% 7-8,20-22h, 1% else), OsVer(95% new, 5% old).

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